Global Manufacturing Company

Enterprise Data Governance Framework Design & Implementation for a Global Manufacturing Company

Presented by **Felice Benita**

Executive Summary (1)

Business Context

The Global Manufacturing Company operates multiple manufacturing plants across different countries, supported by several enterprise platforms including ERP, MES, SCM, PLM, CRM, and IoT systems.
Rapid business growth, acquisitions, and decentralized operations have resulted in fragmented master data, inconsistent reporting, and limited enterprise-wide data visibility.

Current Challenges:

- Multiple ERP instances with inconsistent master data
- Duplicate Material, Vendor, and Customer records
- Low confidence in executive reporting
- Manual reconciliation across plants
- No enterprise-wide Data Ownership
- Limited metadata and data lineage
- Inconsistent Data Quality Rules
- Increasing regulatory and audit requirements

Executive Summary (2)

Enterprise data has become a strategic business asset, yet inconsistent governance limits its value.

The company currently manages data across multiple:

- ➡️ ERP systems
- ➡️ Manufacturing systems
- ➡️ CRM applications
- ➡️ Spreadsheets
- ➡️ Regional databases

The absence of a unified governance model has resulted in:

- Inconsistent master data
- Varying business definitions
- Duplicate records
- Limited data accountability

Key Outcomes:

- ✔️ Standardized enterprise data
- ✔️ Improved data quality
- ✔️ Clear ownership
- ✔️ Faster reporting
- ✔️ Better compliance
- ✔️ Trusted analytics

Business Drivers

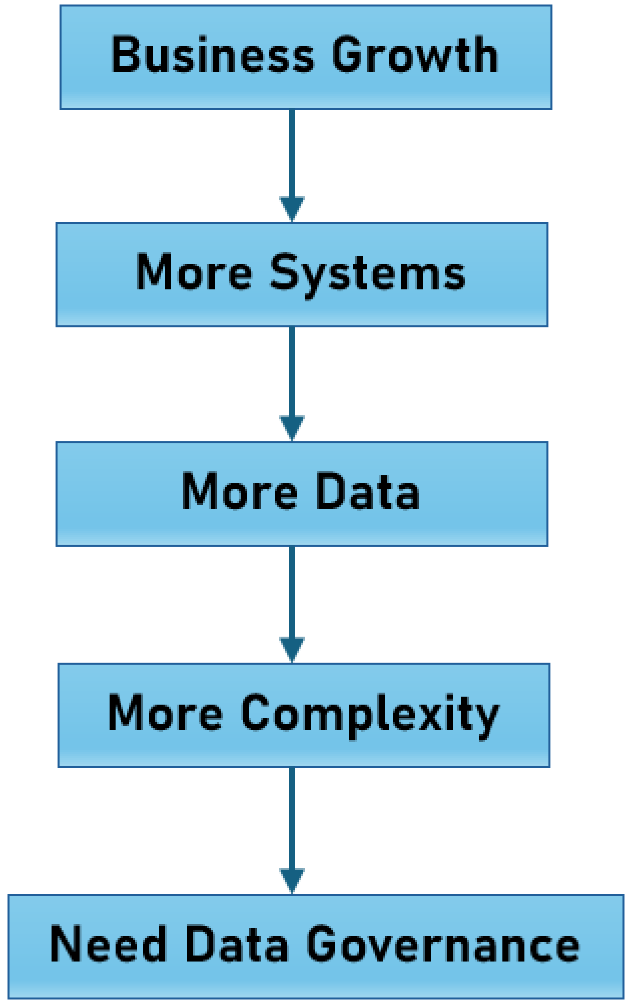
Rapid business growth requires trusted, standardized, and governed enterprise data.

Increasing business complexity has significantly expanded the volume, variety, and velocity of enterprise data.

Key strategic initiatives include:

- Global manufacturing operations
- Multi-country supply chain
- Customer-centric operations
- Digital transformation
- AI & Advanced Analytics
- ESG Reporting
- Regulatory Compliance
- Industry 4.0

Without strong governance, these initiatives face increased operational and compliance risks.



Business Drivers:

- 🏭 Manufacturing
- 🚛 Supply Chain
- 📦 Procurement
- 🌍 Global Expansion
- 📊 Analytics
- 🤖 AI

Transformation Vision

- **Designed a comprehensive Enterprise Data Governance Framework** to establish clear governance structures, standardized policies, and defined data ownership across global manufacturing operations.
- **Strengthened enterprise data management capabilities** through Data Quality Management, Metadata Management, Master Data Management (MDM), Data Security, and Data Lifecycle Management to improve data accuracy, consistency, and compliance.
- **Developed an integrated enterprise data architecture** connecting ERP, MES, SCADA, Industrial IoT, PLM, SCM, WMS, and analytics platforms to enable a trusted **Single Source of Truth (SSOT)** and seamless end-to-end data flow.
- **Defined governance KPIs, success metrics, and a phased implementation roadmap** to measure governance maturity, monitor performance, and support continuous improvement across the organization.
- **Established a scalable, business-driven data foundation** that enhances operational excellence, supports digital manufacturing and Industry 4.0 initiatives, and enables trusted analytics, AI, and data-driven decision-making.

Project Overview (1)

Project Objectives

✓ Establish enterprise-wide data governance

✓ Define a clear **Target Operating Model (TOM)**

✓ Improve data quality

✓ Establish a **Single Source of Truth (SSOT)**

✓ Strengthen **data ownership, stewardship, and accountability**

✓ Enhance **data security, privacy, and regulatory compliance**

✓ Enable seamless integration across **ERP, MES, SCADA, Industrial IoT, PLM, SCM, WMS, and enterprise analytics platforms**

✓ Support **digital manufacturing & Industry 4.0**

✓ Enable AI/Analytics readiness

Scope of Data Governance

Business Functions

- Manufacturing Operations
- Supply Chain & Logistics
- Procurement
- Sales & Customer Management
- Finance & Controlling
- Engineering & Product Lifecycle Management (PLM)
- Quality Management
- Warehouse Management
- Maintenance & Asset Management
- Human Resources

Project Overview (2)

Scope of Data Governance

Core Data Domains

- Material Master
- Product Master
- Bill of Materials (BOM)
- Routing & Production Master
- Supplier Master
- Customer Master
- Plant & Warehouse Master
- Equipment & Asset Master
- Employee Master
- Inventory Data
- Production Data
- Quality Inspection Data
- Maintenance Data
- Financial Data

Scope of Data Governance

Enterprise Systems

- Enterprise Resource Planning (ERP)
- Manufacturing Execution System (MES)
- Supervisory Control and Data Acquisition (SCADA)
- Industrial Internet of Things (IoT)
- Product Lifecycle Management (PLM)
- Supply Chain Management (SCM)
- Warehouse Management System (WMS)
- Customer Relationship Management (CRM)
- Enterprise Data Warehouse (EDW)
- Enterprise Data Lake
- Business Intelligence & Analytics Platforms

Expected Business Outcomes

FROM
Fragmented Data Landscape
↓
TO
Trusted Enterprise Data Platform

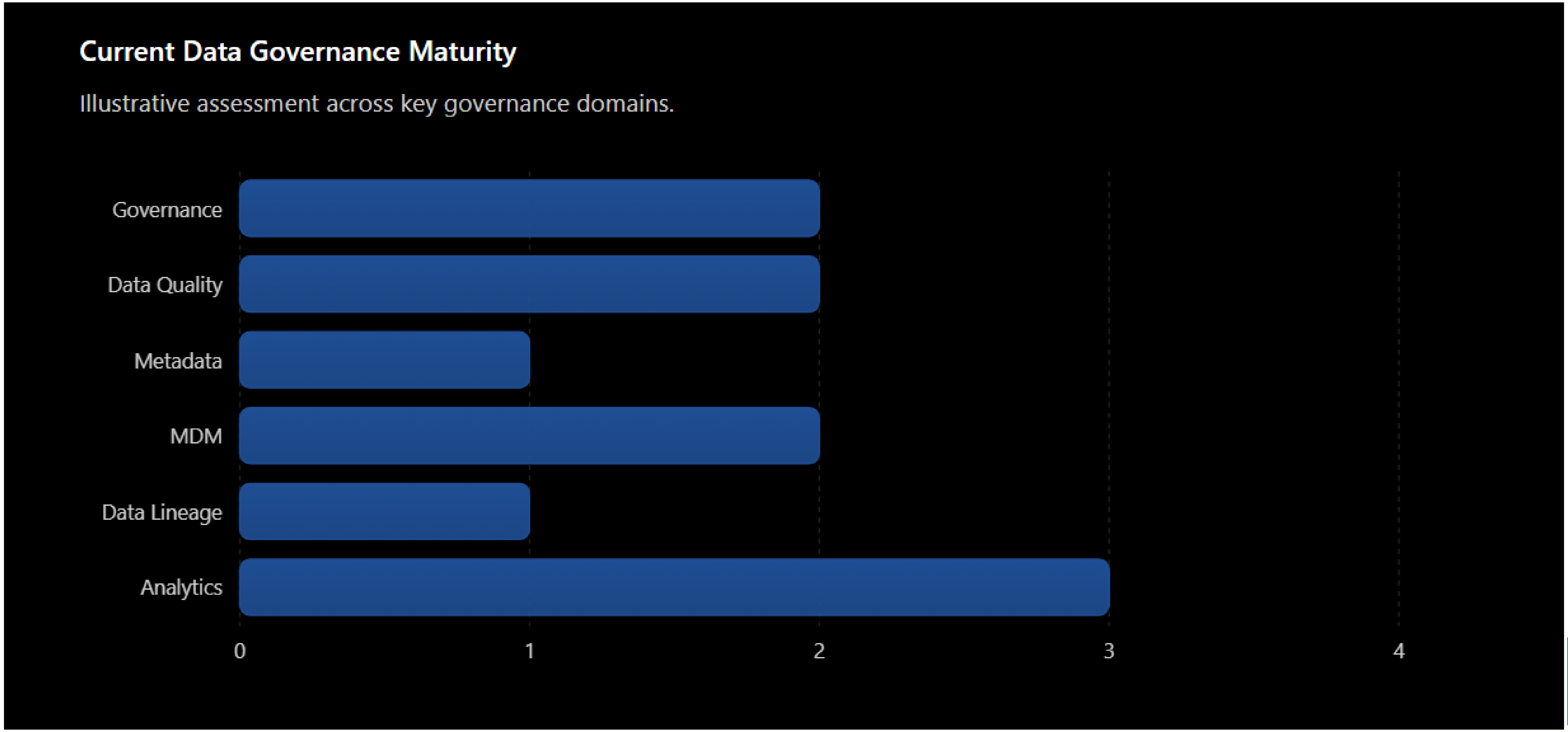
THROUGH
Standardized Governance
↓
High-Quality Data
↓
Global Visibility
↓
AI & Advanced Analytics Ready

Strategic Objective	Expected Benefit
Improve Data Quality	>95% Enterprise Data Quality Score
Increase Accountability	100% Data Ownership Coverage
Reduce Master Data Duplication	>30% Reduction
Improve Reporting	Single Source of Truth
Accelerate Decision-Making	Faster Executive Reporting
Strengthen Compliance	Improved Audit Readiness

Current State Assessment (AS-IS)

Business Area	Current Challenges	Business Impact
Master Data	Duplicate Materials	Inventory Inaccuracy
Manufacturing	Inconsistent BOM	Production Delays
Supply Chain	Duplicate Vendors	Procurement Risk
Sales	Duplicate Customers	Poor Customer Experience
Finance	Different Reporting Logic	Inconsistent Financial Reports
Analytics	Multiple Data Definitions	Low Executive Confidence

Current Data Governance Maturity



Target: 5

Gap Assessment

Gap Analysis Matrix

Capability	Current State	Target State	Priority
Data Governance Organization	Department-based	Enterprise-wide Governance Office	High
Data Ownership	Partial	100% Assigned	High
Data Stewardship	Informal	Formal Steward Network	High
Data Quality	Manual	Automated Monitoring	High
Metadata Management	Spreadsheet-based	Enterprise Data Catalog	High
Master Data Management	Multiple Versions	Golden Record	High
Data Lineage	Limited	End-to-End Lineage	Medium
Governance KPIs	Not Defined	Executive Dashboard	Medium
Data Architecture	Fragmented	Integrated Enterprise Architecture	High

Gap Assessment

Capability Gap Analysis

Domain	Current	Target	Gap
Governance	2/5	5/5	High
Data Quality	2/5	5/5	High
Metadata	1/5	5/5	Very High
MDM	2/5	5/5	High
Data Security	3/5	5/5	Medium
Analytics	3/5	5/5	Medium

Root Causes

Lack of Enterprise Governance

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No Standard Policies

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Inconsistent Business Rules

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Poor Data Quality

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Low Reporting Confidence

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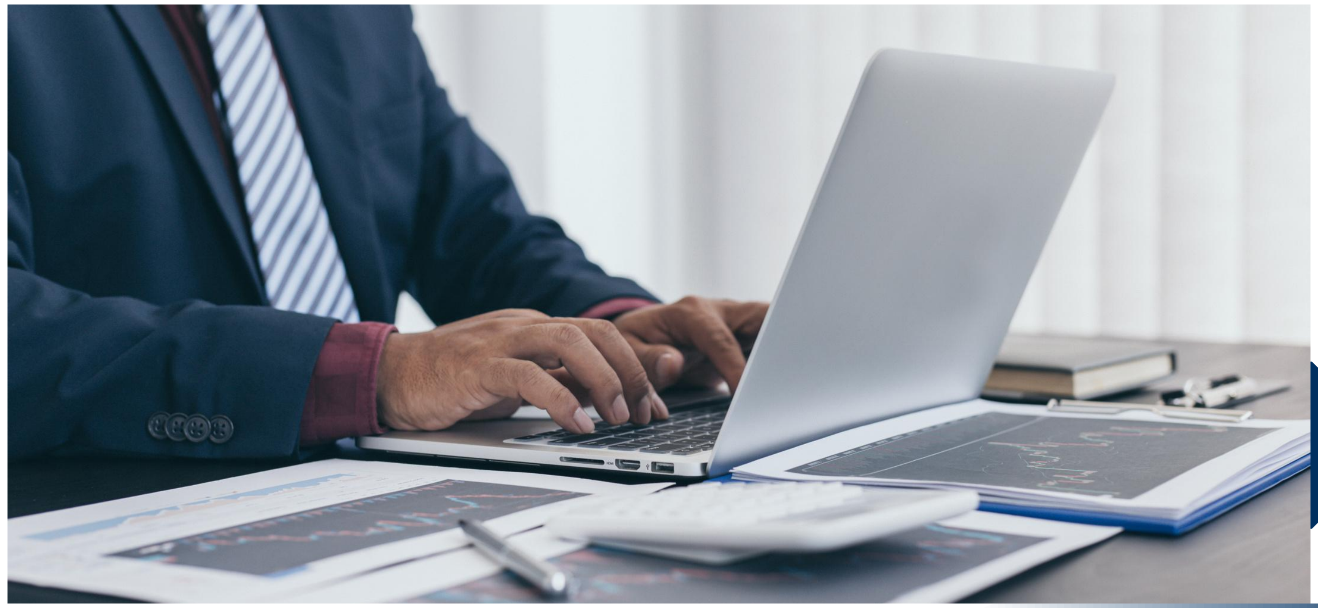
Operational Inefficiency



Target Operating Model (Federated)

Governance Model Type:

Federated Governance Model

Recommended for a Global Manufacturing Company because:

- Global standardization of data policies and definitions
- Local accountability through Data Owners and Data Stewards
- Faster issue resolution within business domains
- Consistent master data across plants and regions
- Better support for mergers, acquisitions, and new manufacturing sites
- Improved data quality and trusted reporting
- Scalable governance as the organization grows

Target Operating Model (Federated)

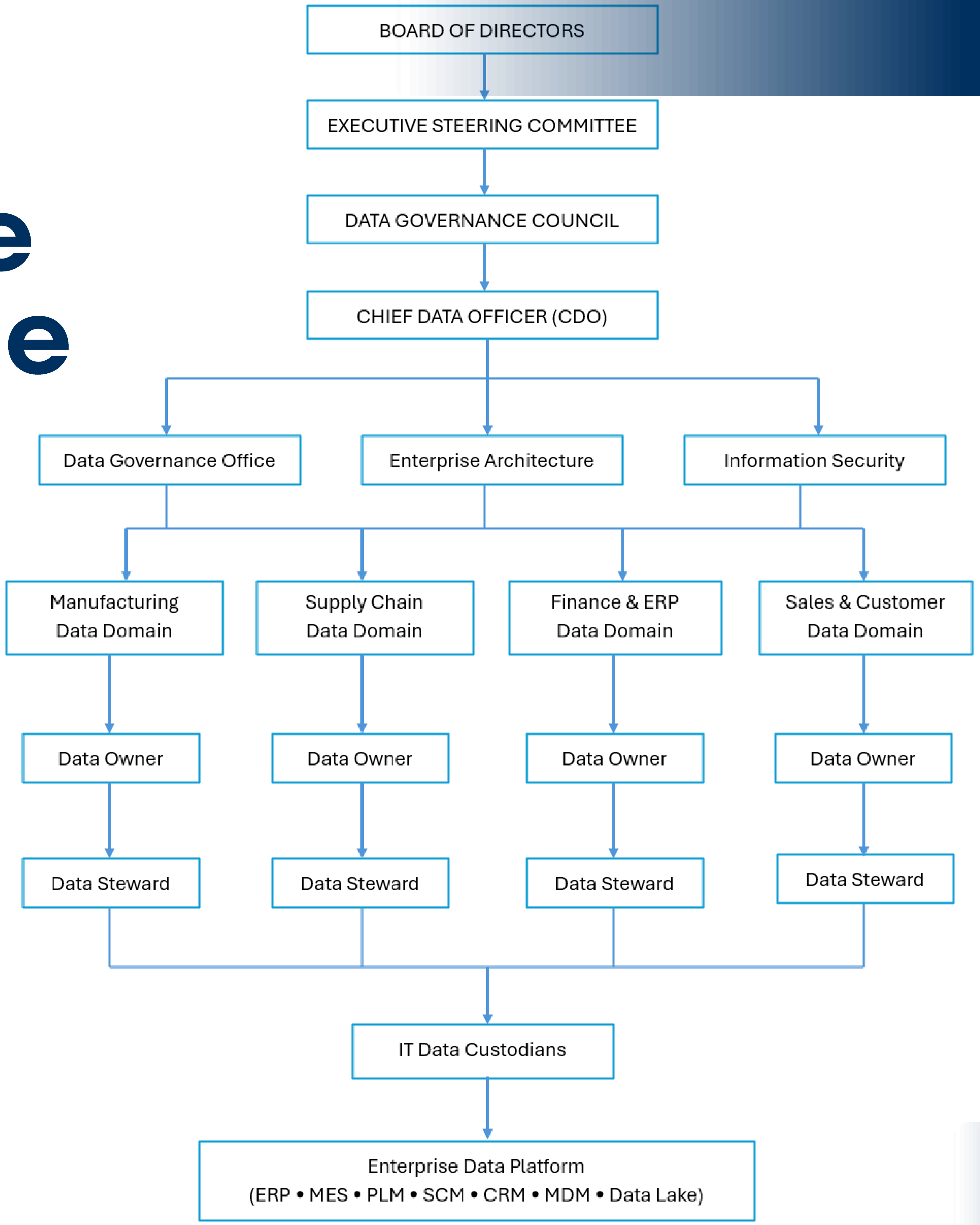
Link to: DATA GOVERNANCE TARGET OPERATING MODEL (TOM)

RACI Matrix

Governance Activity	DGC	CDO	DGO	DO	DS	IT	EA	SEC
Approve Data Governance Policy	A	R	C	I	I	I	I	C
Approve Enterprise Standards	A	R	C	C	I	I	C	C
Define Business Glossary	I	C	R	A	R	I	C	I
Define Critical Data Elements	I	C	R	A	R	I	I	I
Data Quality Rules	I	C	C	A	R	C	I	I
Master Data Governance	I	C	R	A	R	C	C	I
Metadata Management	I	C	A	R	R	C	C	I
Data Quality Monitoring	I	C	C	A	R	C	I	I
Issue Resolution	I	C	C	A	R	C	I	I
KPI Reporting	C	A	R	C	C	I	I	I
Access Approval	I	C	I	A	C	R	I	C
Security Controls	I	I	I	C	I	R	C	A

Legend: R = Responsible, A = Accountable, C = Consulted, I = Informed

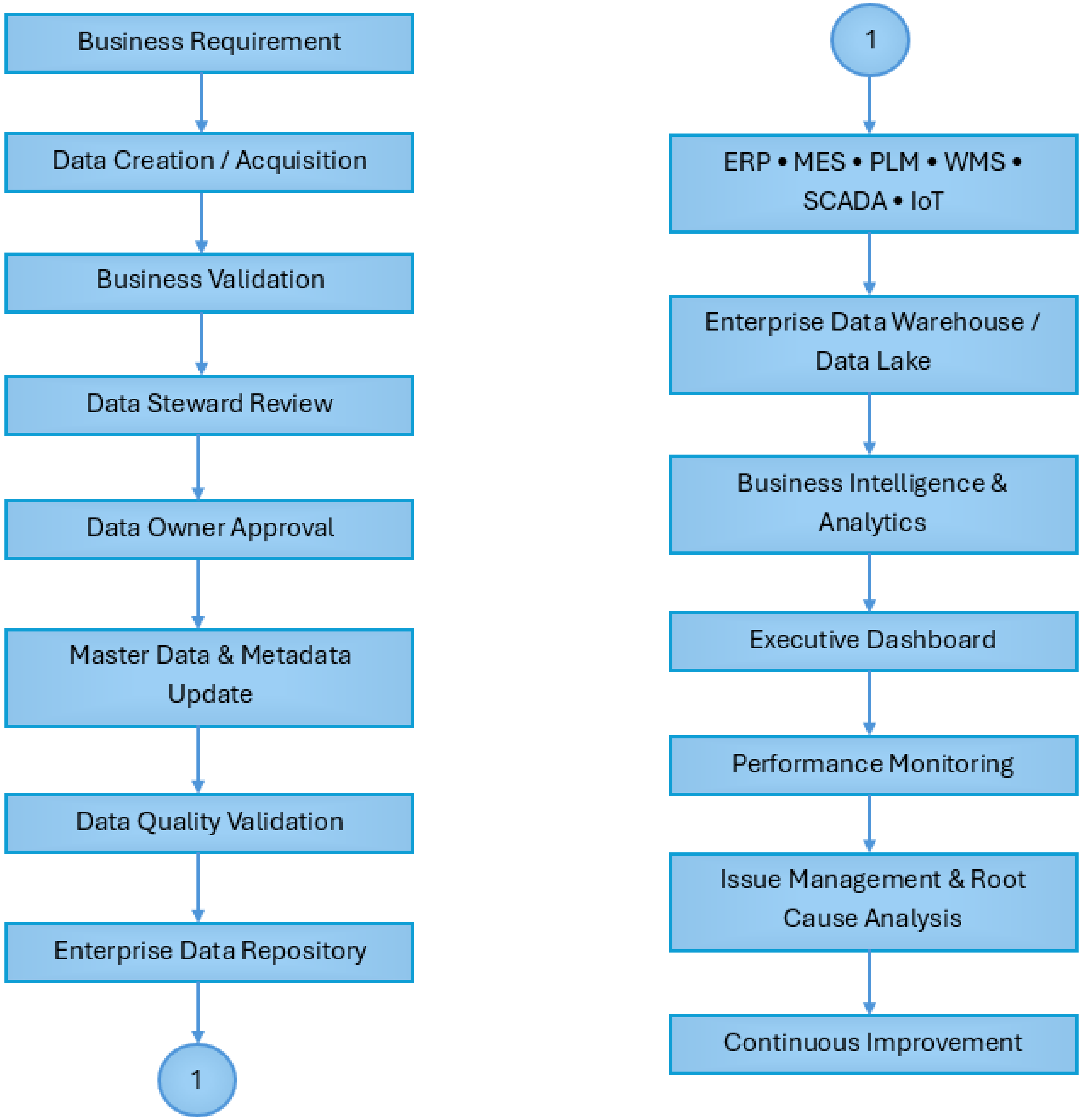
Data Governance Architecture



Zoom Diagram

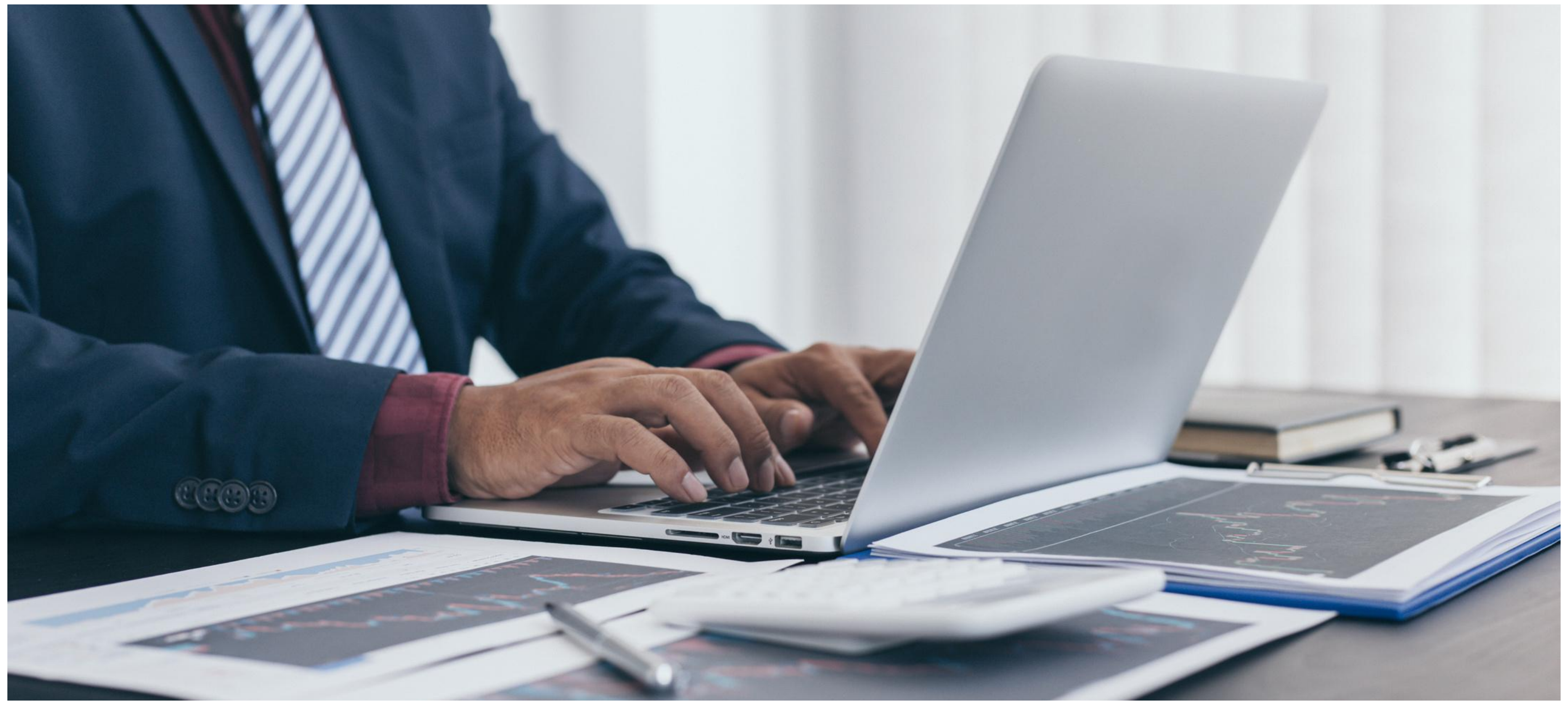
Governance Flow Diagram

Zoom Diagram



Data Governance Policy

Link to: 📜 DATA GOVERNANCE POLICY



Data Quality Management Framework

Vision:

Deliver trusted, accurate, complete, and timely enterprise data to support operational excellence, global manufacturing execution, supply chain optimization, regulatory compliance, and data-driven decision-making.

Framework Overview

Enterprise Data Governance

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Enterprise Data Quality Framework

- Data Standards
- Data Quality Policies
- Data Quality Rules
- Data Profiling
- Data Quality Monitoring
- Issue Management
- Root Cause Analysis
- Corrective Actions
- KPI Dashboard

↓

Trusted Enterprise Manufacturing Data

Data Quality Operating Model

Business Process
↓
Data Creation
↓
Validation Rules
↓
Data Quality Engine
↓
Monitoring
↓
Issue Detection
↓
Steward Review
↓
Remediation
↓
Continuous Improvement

Governance Roles

Role	Responsibility
Data Owner	Accountable for data quality
Data Steward	Daily monitoring and remediation
IT Data Custodian	Technical controls and automation
Data Governance Office	Governance oversight
Data Governance Council	KPI review and strategic decisions

Data Quality Dimensions

Enterprise Standards

Dimension	Manufacturing Example	Target
Accuracy	Correct material specifications	≥98%
Completeness	Mandatory BOM fields populated	≥95%
Consistency	Same product code across ERP and MES	≥98%
Timeliness	Production data updated within SLA	≥95%
Validity	Values comply with business rules	≥99%
Uniqueness	No duplicate supplier records	≥99.5%

Data Quality Rule Library

Data Domain	Example Rule
Material	Material ID must be unique
Product	Product status must reference approved values
BOM	Components must exist in Material Master
Supplier	Supplier tax ID must be unique
Inventory	Stock quantity cannot be negative
Purchase Order	Supplier must exist before PO creation
Equipment	Asset ID must be unique
Production Order	Plant code must be valid

Data Quality Monitoring

Source Systems
↓
Profiling
↓
Quality Rules
↓
Quality Score
↓
Dashboard
↓
Issue Management
↓
Improvement Actions

Implementation Roadmap - Overview

Executive Roadmap Overview

Phase	Timeline	Primary Objective
Phase 1	Months 1–3	Governance Foundation
Phase 2	Months 4–6	Governance Standardization
Phase 3	Months 7–12	Technology Enablement & Enterprise Rollout
Phase 4	Months 13–18	Optimization & Continuous Improvement

Implementation Roadmap (1)

Phase 1

Governance Foundation (Months 1–3)

Deliverables:

- Governance Charter
- Governance Organization
- Operating Model
- Policy Framework
- Governance Roles
- Current State Report
- Gap Assessment
- Data Domain Catalogue

Phase 2

Governance Standardization (Months 4–6)

Deliverables:

- Enterprise Business Glossary
- Metadata Repository
- Data Catalog
- Data Quality Rules
- MDM Standards
- Governance Procedures
- Standard Operating Procedures (SOPs)

Implementation Roadmap (2)

☑ Phase 3

Technology Enablement & Enterprise Rollout (Months 7–12)

Deliverables:

- Production-ready Governance Platform
- Data Catalog
- MDM Hub
- Automated Data Quality Monitoring
- KPI Dashboard
- Governance Training Materials
- Change Management Plan

☑ Phase 4

Optimization & Continuous Improvement (Months 13–18)

Deliverables:

- Governance Performance Dashboard
- Continuous Improvement Register
- Updated Governance Policies
- AI-enabled Data Quality Capabilities
- Governance Maturity Assessment Report
- Executive Performance Report

Enterprise Data Architecture

GLOBAL MANUFACTURING ENTERPRISE

Business Applications

ERP MES PLM SCM CRM HCM
WMS QMS EAM TMS SRM

Operational Technology (OT)

SCADA
PLC
Industrial IoT
Production Machines
Robotics
Sensors

Enterprise Integration Layer

API Gateway
Enterprise Service Bus (ESB)
Event Streaming
ETL / ELT
Data Integration Platform

Enterprise Data Platform

Enterprise Data Lake
Enterprise Data Warehouse
Master Data Management
Metadata Repository
Business Glossary
Data Catalog
Data Quality Engine

Analytics & Consumption

Power BI
Tableau
Executive Dashboard
Self-Service BI
Machine Learning
AI
Predictive Analytics

Architecture Principles:

- One enterprise data platform
- Centralized governance with federated ownership
- Single Source of Truth (SSOT)
- Event-driven integration where appropriate
- Standardized APIs and data contracts
- Metadata-driven architecture
- Security by design
- Scalable cloud-ready foundation

KPIs & Success Metrics (1)

KPI	Before (Current State)	Target (Future State)	Success Criteria
Enterprise Data Quality Score	~78%	**≥95%**	Enterprise data consistently meets quality thresholds across all critical data domains.
Critical Data Elements (CDEs) with Assigned Data Owner	25%	**100%**	Every Critical Data Element has a formally assigned Data Owner accountable for governance.
Business Domains with Assigned Data Steward	30%	**100%**	All business domains have designated Data Stewards actively performing governance activities.
Metadata Coverage	20%	**≥95%**	Critical enterprise data assets are documented in the enterprise metadata repository and data catalog.
Business Glossary Completion	15%	**100%**	Enterprise business terms are standardized, approved, and published across all business functions.
Master Data Golden Record Coverage	45%	**≥98%**	Master data entities (Material, Supplier, Customer, Product, Plant) are managed as Golden Records.
Duplicate Material Records	8%	**<0.5%**	Duplicate material records are minimized through MDM governance and automated matching.
Duplicate Supplier Records	6%	**<0.5%**	Supplier master data duplication is effectively eliminated.
Duplicate Customer Records	7%	**<1%**	Customer master data maintains high uniqueness and consistency.
Data Quality Rule Coverage	35%	**100%**	All Critical Data Elements have documented and implemented Data Quality Rules.
Automated Data Quality Monitoring	20%	**100%**	Enterprise Data Quality monitoring is automated for all critical data domains.
Data Issue SLA Compliance	65%	**≥95%**	Data quality issues are resolved within agreed service-level targets.

KPIs & Success Metrics (2)

KPI	Before (Current State)	Target (Future State)	Success Criteria
Critical Data Issues Open >30 Days	18	**0**	No unresolved critical data issues exceed 30 calendar days.
Policy Compliance Rate	60%	**≥98%**	Business units consistently comply with approved Data Governance Policies.
Quarterly Data Steward Review Completion	40%	**100%**	All scheduled stewardship reviews are completed on time.
Quarterly Data Owner Certification	Not Performed	**100%**	Data Owners formally certify data ownership, quality, and compliance each quarter.
Enterprise Data Catalog Adoption	10%	**≥90%**	Business users actively use the enterprise data catalog to discover and understand data assets.
Data Lineage Coverage	5%	**≥90%**	End-to-end lineage is documented for critical reports and data assets.
Sensitive Data Classification Coverage	40%	**100%**	All sensitive and confidential data assets are classified according to enterprise policy.
Access Review Completion	55%	**100%**	Quarterly access reviews are completed for all governed systems and sensitive data.
High Severity Data Security Incidents	5/year	**0**	No critical data security or privacy incidents attributable to governance failures.
Audit Findings Related to Data Governance	12 Major Findings	**0 Major Findings**	Internal and external audits report no major governance deficiencies.
ERP–MES Master Data Synchronization Success Rate	82%	**≥99%**	Master data synchronization between enterprise systems is reliable and consistent.
Manufacturing Master Data Accuracy	88%	**≥99%**	Material, BOM, Routing, and Plant master data accurately reflect approved business records.

KPIs & Success Metrics (3)

KPI	Before (Current State)	Target (Future State)	Success Criteria
Executive Report Consistency	70%	**100%**	Executive reports are generated from a single governed source with consistent definitions.
Time to Identify Data Issues	5 days	**<4 hours**	Critical data issues are detected rapidly through automated monitoring.
Time to Resolve Critical Data Issues	15 days	**<24 hours**	Critical data issues are resolved within established governance SLAs.
Self-Service Analytics Adoption	20%	**≥80%**	Business users confidently access governed data for analytics without IT intervention.
Business User Trust in Enterprise Data	60%	**≥95%**	Survey results indicate high confidence in enterprise data quality and reliability.
Enterprise Governance Maturity (DAMA/DCAM-aligned)	Level 2 (Developing)	**Level 4 (Managed)**	Governance capabilities are standardized, measured, and continuously improved across the enterprise.

Conclusion & Recommendations

The proposed **Enterprise Data Governance Framework** provides a comprehensive and scalable approach to managing data as a strategic enterprise asset across the Global Manufacturing Company. By establishing clear governance structures, standardized policies, defined data ownership, and integrated data management capabilities, the organization can improve data quality, operational efficiency, regulatory compliance, and decision-making. Combined with a modern enterprise data architecture integrating **ERP, MES, SCADA, and IoT**, the framework creates a trusted data foundation that supports digital transformation, Industry 4.0 initiatives, and future AI-driven manufacturing capabilities.

Global Manufacturing Company

THANK YOU

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